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import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.fft import fft, fftfreq
import os

path = os.path.join(os.path.expanduser("~"),
                    "Desktop", "SE 07 ALML", "PRACTICLE 2",
                    "engine_data.csv")

df = pd.read_csv(path)

print(df.head())

x = df.select_dtypes(include='number').iloc[:,
0].values

fs = 1000
t = np.arange(len(x)) / fs
N = len(x)

f = fftfreq(N, 1/fs)[:N//2]
F = lambda s: 2/N * np.abs(fft(s)[:N//2])

u = x + 0.5 * np.sin(2 * np.pi * 25 * t)
m = x + 0.7 * np.sin(2 * np.pi * 100 * t)

plt.figure(figsize=(9, 6))

plt.subplot(3, 1, 1)
plt.plot(t[:500], x[:500])
plt.title("Time Domain")
plt.grid()

plt.subplot(3, 1, 2)
plt.plot(f[:200], F(x)[:200])
plt.title("Frequency Spectrum")
plt.grid()

plt.subplot(3, 1, 3)
plt.plot(f[:200], F(x)[:200], label="Healthy")
plt.plot(f[:200], F(u)[:200], label="Unbalanced")
plt.plot(f[:200], F(m)[:200], label="Misaligned")
plt.legend()
plt.title("Spectrum Comparison")
plt.grid()

plt.tight_layout()
plt.show()

rms = np.sqrt(np.mean(x**2))

cond = "HEALTHY" if rms < 0.5 else "UNBALANCED" if rms
< 1 else "MISALIGNED/FAULTY"

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print("\nVIBRATION ANALYSIS RESULT")  
print("RMS Value:", round(rms, 4))  
print("Engine Condition:", cond)
```